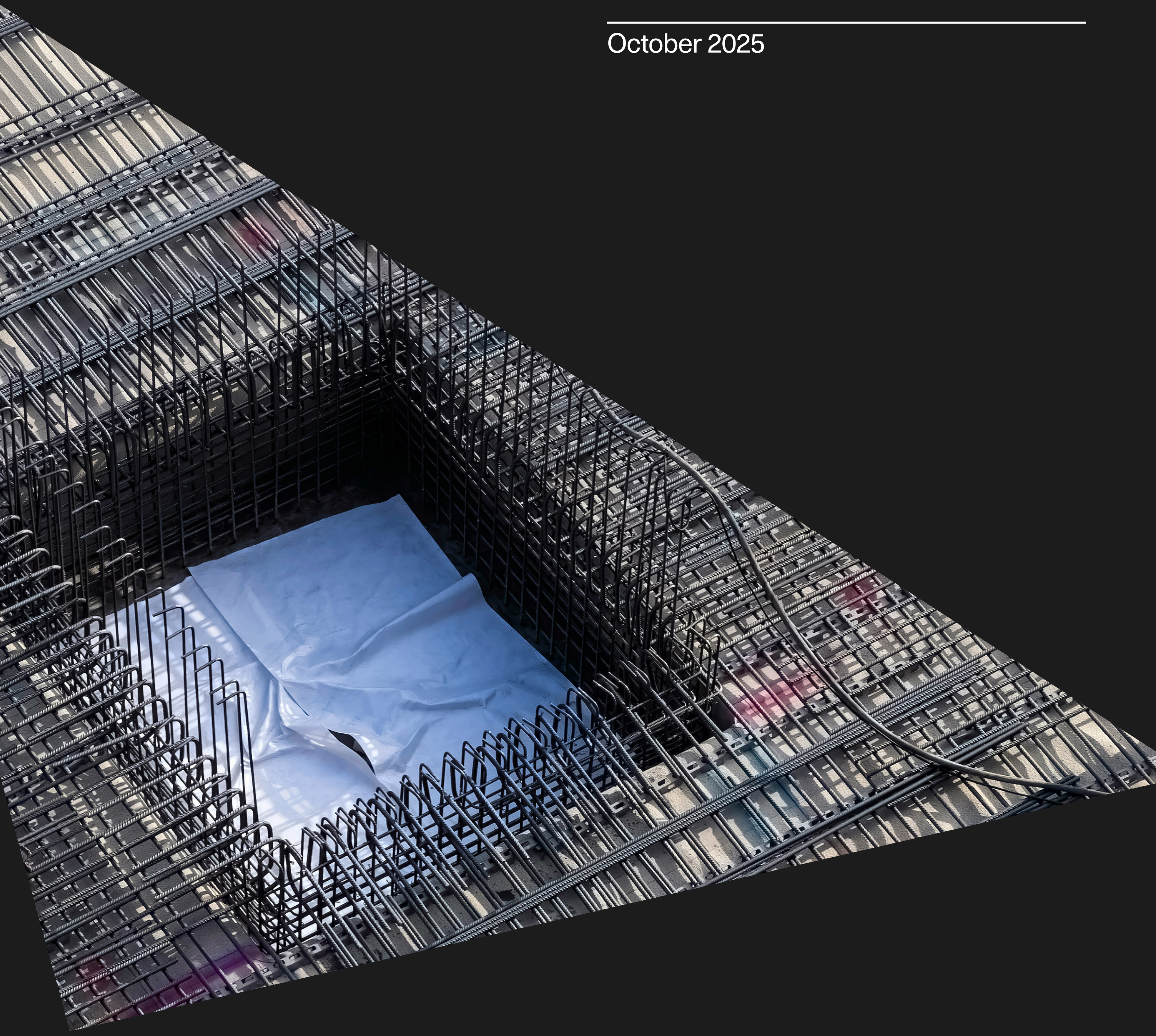
Deep Dive on Data and Information for Critical Infrastructure Management and Maintenance

October 2025



The Alan Turing Institute
LR Foundation
University of Strathclyde Glasgow

Foreword

The scale and pace of infrastructural adaptation and development required to meet the needs of a growing population, and build resilience to unprecedented planetary crises such as climate change, cannot be overstated. According to the United Nations Environment Programme, seventy-five per cent of the infrastructure that will exist by 2050 has yet to be built.

These new infrastructural systems, made increasingly complex and interlinked by advanced digital technologies, need to be properly maintained to ensure those who inspect, manage and operate the assets remain safe, and the supply chains they serve remain unaffected.

Data-driven approaches are the route to ensuring our infrastructure doesn't fail. This report, commissioned by Lloyd's Register Foundation in partnership with The Alan Turing Institute and University of Strathclyde, provides a framework to ensure decision makers in the infrastructure sector are collecting the right data, using it to make evidence-based decisions, and unlock the potential of data-centric engineering for infrastructure management in the United Kingdom.

The Foundation's mission is to engineer a safer world. We hope this report acts as a resource for all those who work in critical infrastructure to adopt new techniques and enable a safer and more sustainable sector.

Ruth Boumphrey
Chief Executive
Lloyd's Register Foundation

Executive summary

This report examines the role of data and information in managing and maintaining critical infrastructure in the UK. It focuses on four sectors with large networks of physical assets: energy, transport, water, and communications. A report from an earlier study into *Critical Infrastructure Management & Maintenance for Safety*, highlighted the need for a deep dive on the role of data and information, posing four questions. These questions acted a catalyst for the study, and were addressed during this research.

This report reviews physical commonalities through categorising critical infrastructure assets into nodes, links, and supporting structures, and highlights the importance of data and records, both analogue and digital, in asset management.

The UK's critical national infrastructure (CNI) sectors share common challenges such as ageing assets, safety concerns, and a skilled workforce shortage. While each sector has its own history and terminology, there are sufficient commonalities to develop transferrable approaches for data collection and analysis. There exists opportunities for innovation in the whole asset lifecycle, from the design phase, through operations and maintenance, upgrades and decommissioning.

While cost and safety drive asset management strategies, sustainability is increasingly desired. How to best incentivise this, when any one individual gain is small, is an open question. Run-to-failure (RTF) is the cheapest asset management strategy but unsuitable for critical assets. Time-based maintenance, driven by regulations and traditional inspection methods, remains prevalent despite aspirations for condition-based methods. Predictive maintenance, encompassing condition-based and risk-based approaches, relies on data analysis to predict asset degradation, but offers the greatest promise.

Leveraging artificial intelligence (AI) will enable degrees of automation in anomaly detection, classification, prediction, and maintenance scheduling tasks. However, challenges exist in balancing data-driven approaches with explainability, aligning rapid technological innovation with slow-moving engineering standards, and ensuring data integrity and risk evaluation for safety-critical applications.

Going forward Standards and ontologies will play a vital role in curating and organising data, enabling both human and AI interpretation. This codification of both explicit and implicit data, information, and knowledge (DI&K) is crucial for implementing AI-enhanced decision-making processes in critical infrastructure. This includes embedding expert knowledge through training programmes and mentoring schemes, as well as leveraging multimodal AI and neuro-symbolic approaches to capture and codify implicit knowledge. While data aggregation is important, the focus should be on data provenance, integrity, and ensuring it represents all expected asset conditions, both healthy and degraded, to avoid bias and data overload.

To realise the vision of a truly data-augmented management of critical national infrastructure, four key areas must be tackled: Improved capture of data and information at point of inspection, Improved management and retention of digital data, information and knowledge, Improve shared understanding of asset degradation, and Maximise utilisation of non-asset management data. We present a roadmap outlining the actions required to address those areas:

Roadmap for data-augmented management of critical national infrastructure

Action: Short term (1-2 years):

Recommendation 1:
Improved capture of data and information at point of inspection

- Asset owners should place increased value on the through-life value of the data associated with their assets. The point of capture offers the greatest opportunity to maximise the translation of the physical observations into digital form.

Recommendation 2:
Improve management and retention of digital data, information and knowledge

- Regulation and standards have a strong role to play here to encourage the adoption and interoperability of data and analytics.
- Create a hub to promote cross-sharing across national initiatives of best practices for ensuring the persistence of data, information, and knowledge.

Recommendation 3:
Improve understanding of asset degradation

- Predictive models are mature, and should be developed and deployed where possible to evidence their suitability and build confidence in their predictions.

Recommendation 4:
Maximise utilisation of non-asset management data

- Asset owners should undertake a review of data gathered for operational purposes for opportunities to leverage this to support detailed, but infrequent inspections
- Exploration of opportunities for automation of multimodal data insight extraction using LLMs

Action: Medium term (3-5 years):

As a sector, more work is required to ensure that explicability of outputs from any data analytic support tool is mandated. Justification of decisions is key, particularly where safety is paramount such as in CNI, and this should be a requirement for any AI-based solution.

Modularise and combine the collection and management of data with a set of useful analytics tools. Use flexible architectures to allow you to start small and scale.

There will be a lag as deployed analytics take time to be used and evaluated, particularly given the long lifetimes of some assets. Building in good asset management practice to the models themselves will provide long term benefits.

Further research is required to build confidence in the use of synthetic data. Demonstration in one safety-critical domain will help build confidence in the application to other related safety-critical domains.

Further research with practical, real world case studies is needed to realise the potential from transfer learning approaches.

The establishment of a sandpit environment to explore and de-risk the implementation of new technologies and approaches in CNI asset management.

Further research is required to better understand the environmental conditions in which CNI will operate in the mid- to long-term future, and what impact this will have on standard asset management practices.

Action: Long term (5+ years):

Scale and maintain data management architectures. This will include introducing relevant training as new systems and approaches are implemented, but also ensuring that there are mechanisms for future proofing and transferring digital knowledge to new implementations of AI systems.

A programme of fundamental research to foresight the next generation of AI-systems and how the lifecycle of the embedded knowledge which supports (and potentially makes) decisions will be managed.

Structure of the report, flow of information, and alternative ways to navigate the content

The report is divided into Context, Main Body and Appendices. The main body comprises five parts:

Context
– About this report
– Foreword
– Executive summary

Main body
– Part 1: Critical infrastructure assets
– Part 2: Asset management strategies
– Part 3: Data and information
– Part 4: Key themes
– Part 5: Conclusions and recommendations
– Bibliography

Appendices
– About the authors
– Workshop participants

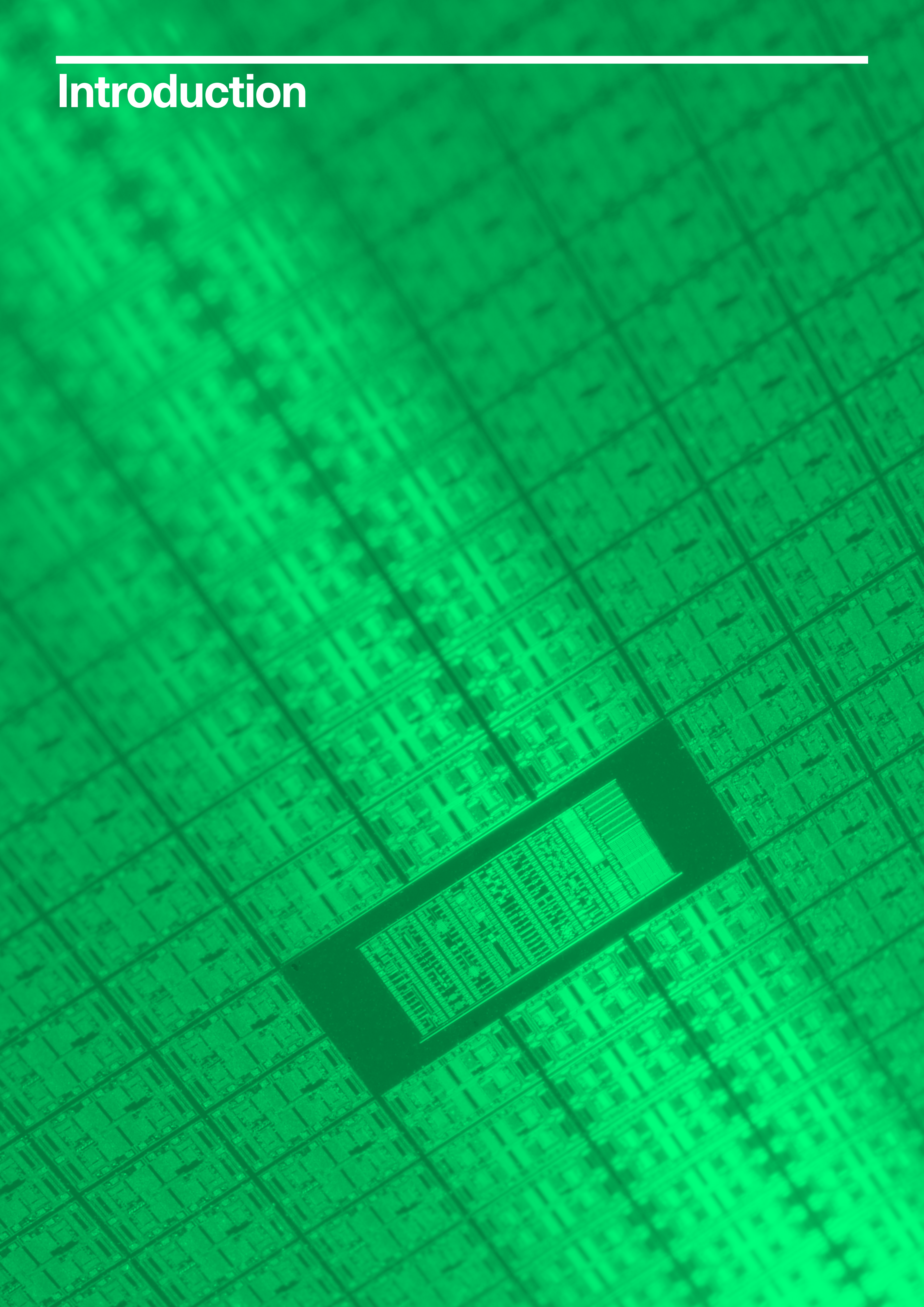
This report follows the same structure as adopted within the *Critical Infrastructure Management & Maintenance for Safety* report which identified the need for this follow-on deep dive into the role of data and information.

There are multiple pathways through the report, depending on how to navigate and absorb the content. Reading the report sequentially, from front to rear cover may, for some, be too much for a single sitting.

The Executive Summary provides a 5 minute overview. Parts 2 and 3 deal with the background into asset management from a physical perspective, setting the scene and context for critical infrastructure asset management in the present day. Part 4 has a more digital focus, looking at tools, techniques and approaches through the use of data science and AI. For a focus only on the outcomes and discussion points see Part 5. For a full absorption, a front to back read will require upwards of a one hour sitting.

Contents

Introduction



Background

The UK's Critical National Infrastructure (CNI) is expected to become increasingly intertwined with advanced digital technologies, as novel methods of collecting and analysing data move from research applications to industrial adoption. This offers prospective benefits to operational efficiency, safety, and sustainability.

However, unlike other technological sectors, traditional engineering disciplines are associated with some unique challenges in adopting these new technologies. Namely, that incorrect or otherwise unhelpful outputs from complex models risk catastrophic consequences for the public. Additionally, there may be scepticism of activities intended to encode knowledge into automated processes that already exists in a skilled workforce which introduces a barrier to adoption. Finally, the maturity of the physical assets, the in-built conservatism of the designs and manufacture of the assets, and a system of knowledge, practice and standardised guidance has grown over decades. This offers a stark contrast to the fast paced, continually evolving digital technologies which are being increasingly deployed to support human decision making.

Aim of report

In 2022 a report was published on a project summarising the evidence and insights on a study into the subject of safety of critical infrastructure [1]. A significant outcome of this report was to recommend a follow-on deep dive focus on the use of data and information for asset management of critical infrastructure. Specifically, four key questions were highlighted in the original report, namely:

1. What level of predictability (for different asset types and systems) do we need in our infrastructure?

2. What does codification of predictability mean for sourcing and prioritising data needed for decision making (rather than the data which is nice to have)?

3. Which data sets, when aggregated, could provide game-changing insights for the design and maintenance of critical infrastructure?

4. What type of safe places and safe mechanisms do we need to liberalise data sharing and minimise data protectionism?

This document reports the outcomes of a project that used these four questions as a starting point for reviewing the use of data and information in the context of the asset management of critical infrastructure. Three workshops were held with participants from across a range of organisations involved with critical infrastructure, and the outputs from these workshops, combined with a desk study constitutes the body of work reported herein. A list of participants is provided in an appendix, but included asset managers, senior policy managers, heads of innovation, data scientists and senior academics drawn from energy, transport, public sector, software industry, general infrastructure and academia. Within this report, we first define what is meant by asset types and systems, while exploring the commonalities of these across the relevant critical infrastructure domains. Approaches to asset management are then outlined and contextualised in the form of current best practice across the CNI sectors. The role of data and information for supporting decision making in critical infrastructure is then examined, followed by a section of key discussion points. The report is finished by providing a response to the original questions posed, followed by a list of recommended future work.

Part 1: Critical infrastructure assets

What is critical infrastructure?

This report focuses on the role of data and information to support the management and maintenance of critical infrastructure. Within the UK, critical infrastructure is defined [2] as:

"Those critical elements of Infrastructure (facilities, systems, sites, property, information, people, networks and processes), the loss or compromise of which would result in major detrimental impact on the availability, delivery or integrity of essential services, leading to severe economic or social consequences or to loss of life."

In the UK, there are 13 critical national infrastructure (CNI) sectors, namely: Chemicals, Civil Nuclear, Communications, Defence, Emergency Services, Energy, Finance, Food, Government, Health, Space, Transport and Water. Four of these sectors, Energy, Transport, Water and Communications, have a large network of physical assets. The management of assets for these four sectors comprises the bulk of the findings in this report, as these shared the greatest commonality in the engineered infrastructure. The energy sector has a large focus on electricity, with oil and gas continuing to play a significant role within sustainability and emissions constraints, with the potential around the hydrogen economy steadily growing. Water, like energy, generally flows from source to consumer, though clean and waste streams are kept separate. Transport and communication infrastructure tend to be more of a mesh configuration, with traffic flowing bi-directionally.

The communication sector is interesting in that a portion of the transmission medium doesn't require a physical link, for example in radiofrequency and satellite communications. Physical connections, such as a country's fibre backbone still play a significant role.

In September 2024, the UK government included Data Centres as part of their critical infrastructure [3]. This highlights the increasing role that data and digital infrastructure are expected to play in supporting the built environment. The announcement places data centres on an equal footing with services like energy and water, with respect to protection from cyber-attacks, environmental disasters, and IT blackouts.

In addition to the engineered CNI, another important aspect is management of the surrounding environment. Broadly speaking this is separated into the natural environment, covering assets like flood plains, wetlands and the subsurface geology in general, and engineered environments such as flood defences, embankments and culverts. While these are managed, they are not primarily used for the fulfilment of the primary function, and therefore not considered to be critical infrastructure.

Each sector has its own domain-specific terminology to classify and categorise the assets. For the purposes of this report, Figure 1 shows the categories used to group the key elements and allow parallels to be drawn across sectors.

Sectors	Assets		
	Nodes	Links	Supporting structures
Energy	Power stations, substations, gas terminals	Gas pipes, overhead lines, underground cables	Steel towers, wooden poles, gantry structures
Water	Waste water treatment plants, pumping stations	Pipes, screens, filters	Culverts, trenches
Communications	Telephone exchanges, data centres	Optical fibre cables, telephone lines	Wooden poles, masts
Transport	Airports, train stations, service stations	Railway track, roads	Tunnels, bridges, gantries, retaining walls

Figure 1: Asset types

Asset types

Assets can be broadly grouped into three main categories, describing the major physical components which comprise critical infrastructure, when it is thought of as a system. These categories are nodes, links and supporting structures.

Nodes

Nodes are used to describe the connection points in the network where two or more links meet. They tend to be a single site which performs some function, beyond enabling the transmission and/or delivery of the service. These nodes tend to represent critical parts of the network and often contain complex subsystems which may offer control or reconfiguration capabilities.

Links

Links are used to describe the physical assets between nodes. These tend to be low complexity and passive in their nature, in the sense that there are no moving or interacting sub-elements.

Supporting structures

Supporting infrastructure is used to describe the assets that maintain the links and nodes fixed position within the environment, and mitigate some of the effects of environmental conditions. These are not primarily responsible for the delivery of the critical infrastructure service or function, and may be shared between services and/or wider land use.

Other assets

While the nodes, links and supporting structures categories captures the bulk of the physical assets, there are a number of other aspects which may be useful in considering. Firstly, there are assets associated with operating and protecting the assets, which includes control rooms, protection equipment, switches and so forth. Much of this functionality will be embedded within equipment contained in the nodes, but some also appears distributed on the network in the links. For example switching points and level crossings on railways or auto-reclosers on overhead electricity lines. Secondly, the asset owner's workforce, and its associated expertise and many years' experience of safe and successful operation and maintenance of the critical infrastructure. Thirdly is the data and records of asset health and utility over its lifetime, which used to justify continued safe operation. There is strong argument that the data itself should be treated as an asset, and the relationships between data, information and knowledge and their corresponding capture, storage and utilisation by both humans and increasingly by computer systems including artificial intelligence, is explored in part 3.

Some examples of detailed descriptions are provided in the following references. The Electronic Communications Resilience and Response Group [4] provides a detailed description of the assets in the telecommunications sector. From the energy sector, The Office of Gas and Electricity Markets (Ofgem) defines asset types under the Network Asset Risk Management (NARM) regulations, and separate out transmission assets [5] and distribution assets [6] separately. Gas transmission and distribution assets are broken down in [7].

1812	National Grid gas started [8]
1865	Joseph Bazalgette’s London sewer network built [9]
1912	Nationalisation of UK telephones under the Telephone Transfer Act 1911 [10]
1926	Electricity (Supply) Act 1926 recommends that a “national gridiron” supply system be created [11]
1945	Nationalisation of water services and establishment of regional water authorities under Water Act of 1945 [9]
1954	UKAEA created [12]
1956	Calder Hall opened by the Queen [12]
1972	Power shortages following miners strikes [13]
1986	Privatisation of British gas
2002	Network Rail established [14]
2003	London blackout affecting 500,000 people [13]
2003	Blyth offshore wind farm (UK’s first, and then largest in the world) begins operation [15]
2015	Highways England established [16]
2024	Data centres added to UK CNI [3]

Timeline

A common theme in the selected key historical dates for UK CNI is the establishment of organisations providing regulatory and technical oversight, which has led to development of standards and adoption of safe practices.

Commonalities and interfaces

Identifying commonalities in the characteristics of the physical assets, allows generalities to be drawn about how transferrable approaches to gathering, interpreting and managing the associated data might be adopted. Areas of commonality include:

Age of assets: All the assets are of a mixed vintage, but all 4 sectors are at least 75 years old. While not all the assets are of that vintage, much of the original planning and design of the network routes have been in place for this length of time. Technology has evolved across all these sectors, but in general all have resulted in the phasing out and replacing old technology with newer technology, rather than wiping the slate clean. Therefore, a shared challenge for all sectors is that part of the network(s), the assets and the start of life data for parts of the network, do not exist in digital form, if even at all. Furthermore, the CNI systems comprise a range of technologies at a mixture of vintages making standardisation across the entire system extremely difficult.

Safety: A key aspect of all CNI is that of safety, both to the public and to the environment. Much of CNI traverses publicly accessible ground, and therefore failure of part of the system, could result in injury or loss of life through contact with high voltages, road or rail crashes, ruptures of pressurised equipment or flooding. This in addition to the loss of the basic services the CNI provides for society to function. As a result, CNI tends to be a conservative industry which can make the adoption of new technology slow.

Size and complexity: The sheer size and complexity of all the critical infrastructure systems presents common challenges. Gathering data from a wide range of geographically dispersed (but to some extent inter-dependent) assets in a consistent manner, in a range of operating conditions in both urban and remote areas, poses problems.

Ageing workforce: A common challenge is attracting and retaining a suitably skilled workforce in asset management. It requires a high degree of technical knowledge and understanding of how assets degrade and fail under different operating and environmental conditions. Increasingly, strong digital skills are required to interpret the ever-increasing volumes of data it is now possible to gather, and while automation has allowed engineers and asset managers to oversee great parts of the CNI networks, this has focused the collective organisational knowledge into fewer and fewer domain specialists. The skills required by the engineers of the future are laid out in a recent report published by the Royal Academy of Engineering [17]
.

Environment: Another shared common interface is the environment in which the assets reside. This is particularly apparent in the subsurface, where services from several different sectors must co-exist. This is also presenting itself as an opportunity, and initiatives like the National Underground Asset Register (NUAR) is making inroads into opening up the sharing of data in the space across several sectors [18].

While any single CNI sector is complex enough in its own right, the increase in digitalisation is bringing the individual sectors closer together. Electricity networks are reliant on communication between protection devices to ensure they operate correctly and safely during fault conditions. The water sector is becoming more reliant on the electricity sector to power its pumps and control room equipment. These new interdependencies, that are introduced to already complex network structures, will require more careful analysis to ensure 'fail-safe' conditions are met so that component failures do not propagate through the system and result in loss of supply to large populations.

Section summary

This section establishes that while each sector of CNI has domain-specific history, terminology and requirements, there are sufficient commonalities to develop transferrable approaches for data collection and analysis to support safety and sustainability throughout the asset management lifecycle.

The asset management lifecycle of CNI must manage the complexity of system-level and component-level outcomes. Different sectors represent a mixture in the sophistication of available technologies and data, which have evolved through decades of incremental changes. The practical reality is that each CNI sector now has its own standards and legacy constraints, in response to the high consequence of failures to assets that serve the national populations. This represents a barrier to implementing new approaches, which may promise to benefit sustainability and economy.

Part 2:
Asset management strategies

Overview

Asset management is the practice of overseeing assets to realise their maximum value and ensure that they fulfil their function both safely and as intended. For the purposes of this report, this is considered to cover the entire lifecycle. While the value and function of the asset, and the metrics used to quantify the value and performance, changes from domain to domain, the strategies employed are common.

Asset management can be characterised as a series of decisions (opportunities to intervene, or not, in the state of CNI systems), governed by engineers to maximise the value of an asset, while ensuring that it can continue to perform its function without compromising the safety and integrity of the users of the system, the surrounding environment, and the system as a whole. These decisions are made by individuals on behalf of asset owners and operators and are based on information, data and engineering knowledge.

Common practice is to have teams of asset managers and domain specialists' input to decision-making, each with expertise in particular aspects of the asset's health. This ranges from the maintenance engineer who has first-hand knowledge of the current condition of the asset they are maintaining, via their direct observations, through to system operators who understand how the asset's condition is contributing to the system performance, to the manager making strategic decisions about where and when to replace, reinforce or upgrade parts of the system. Data has played a significant role in supporting and justifying decision making, historically through engineer's hand-written notes and logs, recordings from analogue instrumentation, providing a snapshot of asset condition or performance. With the advancement of digital technology, this has enabled the opportunity for more and more automation, including the possibility to automate aspects of decision-making in asset management. While critical infrastructure domains may be conservative, there is a significant track record of automated decision making in certain aspects of operational decision-making in an engineering context. Protection and control of electrical grids is one such example, where protection schemes are designed to make decisions to isolate parts of the network should the presence of a fault be detected. The need for autonomy in this case is driven by the very short time scales over which the decision is needed.

Undoubtedly the decision-making process which this functionality embodies, mimics human reasoning in some aspects (i.e. given a set of conditions, perform a set of actions). This reasoning is hard-wired into the system and closely coupled to the sensor data used to make the same, repeatable decision in a justifiable manner. Indeed, this is reflective of the journey of AI research, where in the 1980's expert systems which embodied this type of reasoning approach were regarded as the cutting edge of AI and found much success. These approaches fell out of favour due to their limitations: the rules they contained were brittle and did not cope well with uncertainty; there was a high cost of developing the knowledge bases in the first place; and although they found their place in automating some decisions, they were not broadly applicable in the eyes of the general public (and funders) in terms of tackling the concept of Artificial General Intelligence (AGI). More recently, methods of mathematical programming and reinforcement learning have emerged as applications of decision optimisation [19]. This illustrates the importance of understanding the role that data, information and knowledge plays across the full asset lifecycle.

The asset lifecycle

The key stages of an asset's lifecycle are illustrated in figure 2. Though this is a simplistic view it provides a scaffold to explore where data is used.

While this approach is useful when describing a single asset, even a complex one such as a nuclear power station or a waste water treatment plant, for CNI there is the additional challenge that the fundamental design and implementation of the backbone has already been done, and the system as a whole has evolved over many years. While the physical assets themselves may not have evolved, the means by which data is captured, processed and stored has, along with the accumulated knowledge and experience of how the assets degrade over time.

Much of asset management relies on experience of how assets have aged under certain conditions, then using this accumulated knowledge to be able to determine how new assets may age in the future.

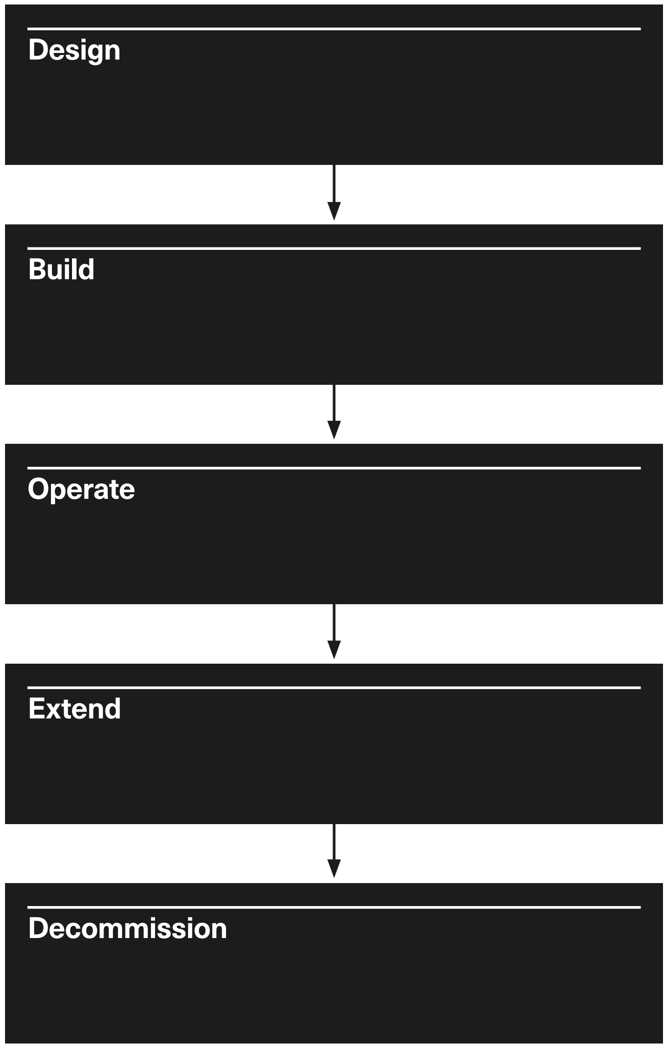


Figure 2: The asset lifecycle

Asset management strategies

The choice of asset management approach is currently driven by two main factors, cost and safety, which can be combined (and traded-off) using the concept of risk as an expected outcome. More recently there is a desire to include a third driver – sustainability - into consideration. While cost and safety have real impetus behind them, either due to economic viability, or legal consequences in terms of safety, sustainability is mainly driven by government policy. Interim targets are set, but often missed without significant consequences. Where tensions exist between the three drivers, it is generally sustainability that gives way.

Run-to-failure

A run-to-failure (RTF) based approach is the cheapest to implement in terms of data and information gathering, but should only be applied to assets where the consequence of failure is small, and replacement of the asset is straightforward. This approach ensures that the maximum value is extracted from the asset, and it is not replaced while it still has the ability to perform its function. However, should the failure of the asset lead to a cascade of failures which affects the wider system, or the ability to replace the asset in a timely manner is not possible, then this choice of strategy is unsuitable.

Current practice

Workshop participants noted that lower-criticality equipment, such as standard air-conditioning units or secondary telecom cabinets, often remain unmonitored until breakdown, simply because failures carry minimal disruption and repair or replacement costs are low. Similarly, long runs of underground cable or minor distribution feeders tend to stay on an RTF strategy. For example, deploying sensors along dozens of kilometers of buried lines is both technically complex and expensive, and some operators preferred to reserve real-time monitoring for higher-voltage, higher-risk circuits that sit “upstream” in the network.

References to an “anti-fragile” ethos were found, contrasting with the “fail-safe” approaches more prevalent in industries such as aerospace. The anti-fragile framing favours over-engineering against shocks, rather than leaning towards continuous, data-driven maintenance. In these settings, upfront CAPEX (Capital Expenditure) investments in robust design still outweigh the OPEX (Operational Expenditure) costs of sensor networks, and teams often trust proven mechanical redundancy over analytics. Some participants highlighted that this historical inertia traces back to RTF practices first formalised in the 1930s and ’40s, which means that asset failure often remains the trigger for intervention, even when engineers can monitor dozens of sites from a single control room.

Illustrative examples of run-to-failure identified include:

Bridges: Although fibre-optic sensor networks exist, many bridges at risk of collapse still rely on human visual inspection because of a lack of trust in automatically collected data. One operator explained that 100 km of fibre-optic cable are already installed on pipelines but noted that, without confidence in the data, teams revert to in-person checks.

Offshore wind turbine foundations (monopiles): Foundations cannot be replaced without decommissioning the turbine, so most owners adopt a hybrid RTF/periodic approach: periodic lifetime-extension surveys but otherwise only intervene upon visible degradation.

Cable jointing: Responsibility for cable joints typically falls on whichever maintenance crew is local to one side of the joint—if the joint fails, someone must attend, but until then it is left unmonitored.

Low-voltage distribution networks: No condition data is gathered unless a fuse blows multiple times; technicians then bring portable testers (providing only a binary health signal) to verify a problem.

Underground infrastructure: The most common approach remains visual inspection, often performed by train drivers or field engineers, before any instrumentation. Where sensors do exist (e.g. for groundwater pressure or vegetation encroachment), data are collected manually and only used when an asset is pre-categorised as high risk.

Time-based

Also sometimes called periodic maintenance, assets are examined at defined intervals to evaluate their condition, and a decision is made to undertake an intervention on the asset or replace it. For some assets, they are replaced regardless of their condition. In order to support the decision-making process, the expected functional lifetime of an asset should be known in advance and used to determine the periodicity of the check or replacement.

Current practice

Alongside RTF, time-based maintenance remains a key approach to asset management. While most organisations aspire to shift to condition-based or predictive methods, several factors were observed that motivate the prevalence of periodic maintenance:

1) Regulatory and warranty drivers:
Many assets are governed by strict regulatory inspection schedules (such as substation safety audits) or by manufacturer warranty conditions that mandate inspections at fixed intervals. The predictability and simplicity of periodic visits is perceived as beneficial by asset managers, not only to maintain compliance, but also to coordinate activities such as renewal works or outage projects. For example, ASME BPV Section XI [20] mandates in-service inspections of Class 1 reactor components every ten years, and the US's Pipeline and Hazardous Materials Safety Administration (PHMSA) would require inline inspection of high-consequence gas pipelines at least every seven years [21].

2) Prevalence of traditional and intuitive inspection practices: Periodic site visits still use traditional and intuitive hands-on inspection methods. For instance, soapy water is applied to gas switchgear to identify leaks, and wooden electricity poles are tapped with hammers by an experienced engineer to ascertain the health of the pole. Companies are conducting efforts to standardise these tests, such as annual refresher training for staff, and organisations are moving away from paper records to a digital form completed by staff in the field on tablets. Participants stressed the importance of training, as these forms use categories for asset health on a scale from 'acceptable' to 'severe'. A common language and understanding of the categories across all staff is essential. The importance of training and learning from experienced staff was further illustrated by the fact that experienced staff can 'intuit' (for example, they have learned the sound of a healthy pole through experience, the same way a doctor learns how to differentiate diagnoses using a tool like a stethoscope).

3) Separation between asset management data and operational data: Operational data is considered separately from asset management data in some organisations. The only data the asset managers 'own' is static data such as the outcome of inspections, and the maximum load across the last twelve months. The frequency of inspections is dependent on the asset and can range from monthly inspections to once every six years. The operations team does collect (and act on) information such as gas analysis from transformers half hourly, but this is not viewed by or used by the asset management department.

Illustrative examples of periodic maintenance identified include:

Wind turbine mechanical components, such as gears, bearings, and shafts: Inspection schedules are primarily defined by contract or warranty rather than by usage metrics.

Transformers: Oil is sampled every three years per manufacturer guidelines. Defective oil may be exchanged on-site, but continuous monitoring is uncommon unless the owner agrees to contribute the additional cost.

Railway track and civil structures: Roughly annual visual inspections form the backbone of maintenance strategy, with evolving scoring systems to reduce subjective variability across inspectors.

Signalling and control systems in railway infrastructure: Time-based safety checks remain standard, though many operators are beginning to migrate data to handheld-enabled asset-management platforms.

Predictive

Predictive maintenance also encompasses condition-based maintenance (CBM), reliability centred maintenance (RCM) and risk-based maintenance (RBM) as related terminology. This strategy relies on being able to gather testing, inspection and/or monitoring data which reflects the condition of the asset. This is analysed using a model of asset degradation, ideally in a variety of environmental conditions and operating regimes. This approach is sometimes described as requiring significant volumes of appropriate data, coupled with a full understanding of possible failure modes and how they manifest in data. However, in principle, risk-optimal data collection is attainable based on any state of knowledge, and methods from computational statistics such as value of information analysis can provide a formal, quantitative method for achieving this [19].

Current practice

Despite the dominance of RTF and periodic regimes, CBM and RBM have become increasingly commonplace, most notably in newer or data-rich industries, with wind power often cited as a leading example. The analysis revealed some key themes applicable across CNI sectors:

1) Drivers of CBM adoption:
In the wind sector, contractual service-level agreements and industry standards explicitly require data collection, anomaly detection, and periodic inspections. CBM is frequently offered "as a service," bringing together vibration and temperature monitoring hardware with analytics and expert interpretation.

2) Data-driven vs. physics-driven decisions: Most CBM implementations rely on data-driven flagging, such as statistical process control, threshold-based alarms, and anomaly detection, rather than detailed physics-based modelling. Setting appropriate thresholds requires domain expertise: false positives are common, and true-failure events are rare, making model validation a continual challenge.

3) Operational workflow: Typically, an asset owner installs sensors and configures alarms: when an anomaly is flagged, alerts flow to both owner and operator. The operator filters out irrelevant or out-of-scope alarms, assesses the system-wide impact, and triggers a maintenance intervention if warranted.

In addition to the above, the following challenges to scaling up CBM approaches in critical Infrastructure were identified:

4) Data quality and accessibility:
Inspection reports often exist in human-readable form (as electronic or paper documents), rather than machine-readable databases, limiting automated analytics.

5) Regulatory conservatism:
Prescriptive standards encourage time-based inspections, and companies fear liability if they stray into purely data-driven regimes. Note that some sectors have standardised guidance on the acceptable use of CBM/RBM.

6) Technical complexity: Long-lived assets (such as water mains and pipelines) may outlast multiple engineering careers, complicating data ownership and model continuity.

7) Bureaucracy and permissions: Even autonomous drone inspections require regulatory clearance, which is an onerous step that is hard to justify for low-consequence assets, such as wooden poles.

Illustrative examples of CBM identified include:

Wind turbine mechanical components: Reliability engineers monitor vibration and temperature channels, using traffic-light schemes to flag deviations. Although periodic inspections remain in the contract, CBM is used to minimize unplanned failures.

Rail networks: Track health is still mostly visual, but risk ranking (1-4) and failure-rate analysis are guiding pilots of sensor-based monitoring for the most critical 2–3 km segments of network.

Vegetation management: Helicopters — or, in limited winter windows, drones — scan for tree encroachment on electrical overhead lines; however, seasonal foliage and required clearances often frustrate continuous aerial inspection.

Prospective benefits of progressing existing practice

The development of sensing technologies, such as those employed by structural health monitoring systems [22] and modelling software with capabilities to assess data at that scale, can offer new insights and facilitate more proactive asset management. Also, new innovation (such as more sophisticated modelling) may be required to address the challenges associated with safety and sustainability of ageing infrastructure.

Novel approaches offer prospective benefits throughout the asset lifecycle: reducing the need of expensive and resource-intensive physical testing during design, by adopting hybrid certification by analysis workflows. Improved quality control analytics may improve the efficiency of construction projects, for example, improved predictive maintenance can reduce downtime and optimise resource allocation during operation, as well as supporting decisions regarding life extension and decommissioning.

Improved data analytics may stimulate innovation in unexpected ways too. Engineering teams may be able to identify new inefficiencies or risks when presented with higher volumes of asset-specific and external data (such as satellite imagery [23]).

Recent examples include: Google DeepMind models used to improve efficiency of wind turbines [24] or UK water companies analysing satellite images [25] and acoustic emission data [26] to repair leaks earlier, saving water and preventing further damage. There are believed to be similar operational optimisation opportunities in the energy industry [27], where forecasting models may help the National Energy System Operator (NESO) more effectively balance energy supply and demand, reducing the risk of requiring energy backup power plants, with higher associated costs and emissions. In 2022, The Alan Turing Institute, in collaboration with Lloyd's Register, completed the first independent verification of a digital twin in the maritime industry [28], which allowed for additional remote testing, offering potential benefits in safety and economy.

Associated risks of implementing novel methods

Perhaps the most impactful result of machine learning approaches in recent years is the availability of generative models, such as Large Language Models (LLMs) for general public use. These neural networks used internet data (with unclear copyright implications), as well as complimentary reinforcement learning, to fit hundreds of billions (and growing) of parameter values, and very impressive outputs were presented. Multi-modal models can extract information from images, agentic systems can perform detailed internet research, and generated text can be indistinguishable from that written by humans.

There have been various newsworthy examples of erroneous outputs from these systems being used by individuals [29] and organisations [30], indicating a lack of awareness of this risk, or having insufficient mitigation in place. Outputs from LLMs are typically accompanied by a message, along the lines of:

'Model X can make mistakes, so you are asked to double-check all outputs'

The above disclaimer would be an unacceptable clause if it were to accompany any engineering analysis. Engineers will not require examples showing the ceiling of proposed complex models, rather they will need to quantify their reliability, i.e. understand the probabilities, however small, to determine whether the output is helpful. This will allow expected consequences to be compared with existing practices, which will also have non-zero risk, so that the appropriate use cases – when their benefits are expected to exceed their risk – can be identified.

Deviating from established and standardised practices also introduces new risk, and workflows that rely on complex models will need to be critically evaluated in this respect. This is similar to how significant changes to a design of a chemical process plant or nuclear facility would trigger comprehensive HAZOP (Hazard and Operability) studies.

Considering some challenges that this would introduce to asset owners:

1) Data management practices, including storage and retention: For data being collected at scale for new applications, new software and competencies may need to be brought in.

2) Data quality: Engineering data can typically involve indirect measurements of complex physical phenomena in challenging environments. Consequently, it will always be (to some extent): imprecise, unreliable, and incomplete. Regression models will suffer from using poor quality data, although probabilistic approaches can directly account for these features (provided they have been quantified).

3) Data sharing: Access to larger datasets, across a broader range of assets can facilitate improved modelling at multiple levels/hierarchies. Industrial data sharing projects must contend with technical, and commercial opposition.

Other, perhaps less tangible or familiar, challenges that have been identified include:

1) Cybersecurity: Introducing a reliance of critical infrastructure on digital controls introduces the risk of disruption by software bugs, introduced either accidentally or targeted by hackers. The EU's AI act [31] lists infrastructure that supplies water and energy as 'high-risk'. Human-in-the-loop systems, with fail-safe conditions may help mitigate these risks. Furthermore, as CNI becomes more digital, this increases the risk of cyber-based incidents. While this is of primary concern to CNI operation, rather than asset management explicitly, the underpinning sensor and digital technology is being used increasingly to support both operation and maintenance.

2) Privacy and ethics: Forecasting models may benefit from collecting data at household or individual level, and regulators will be responsible for defining the reasonable boundaries for such activities, to protect the privacy of the public. Similarly, decisions supported by complex models may begin to disproportionately inconvenience or harm certain regions. Transparent policies and workflows, along with monitoring of model inputs and outputs, may help mitigate this risk.

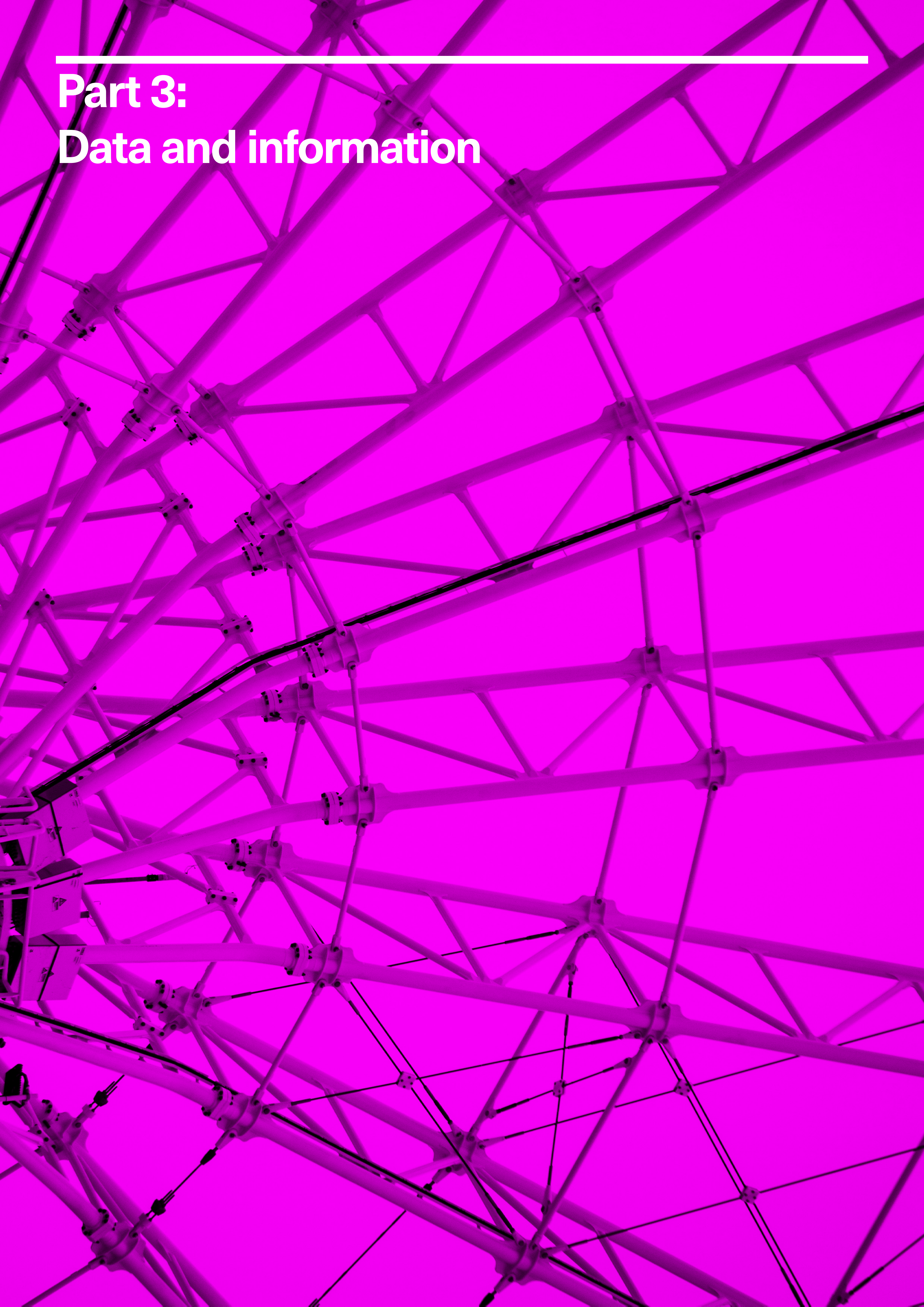
Section summary

Asset management involves overseeing assets to maximise their value and ensure safe, intended function throughout their lifecycle. This process involves decisions made by asset managers and domain specialists, informed by data, information, and engineering knowledge. Common strategies include run-to-failure, condition-based maintenance, and predictive maintenance, each with its own cost, safety, and sustainability considerations.

Condition-based maintenance (CBM) and risk-based maintenance (RBM) are becoming more common in critical infrastructure. CBM relies on sensor data and models to predict asset degradation, but challenges include data quality, accessibility, and the need for domain expertise to set appropriate thresholds. Despite these challenges, CBM is increasingly offered as a service, combining hardware, analytics, and expert interpretation.

Sensing technologies and modern data analytics offer significant potential benefits, such as reduced downtime and optimised resource allocation. However, they also introduce new risks around cybersecurity, data management, regulatory compliance and model risk. Applications of complex models for CNI asset management require careful evaluation against existing practices to identify appropriate use cases.

Part 3:
Data and information



Both data and information play a significant role in asset management. Data is the record of the observations made of the asset at a particular point in time, and information is the context around the quantity. Data and information can take many forms, from temperature sensor readings from underground cables, to images of corrosion on steel towers, to the report of a burst water main.

With the advent of digital technologies, it has become easier and more commonplace to store the data digitally, ultimately encoded as a series of 1's and 0's. Information can be coded digitally too, in the form of data structures and common languages which allow real world concepts to be articulated in a digital form, which allows humans to meaningfully use the digitised data to support decision making.

Indeed, this data and information can be utilised to draw meaningful conclusions, or initiate some corrective action. When these actions are encoded in the individuals' and organisation's memory such that it can be reutilised in the future, this can be considered as knowledge.

Traditionally, this utilisation of knowledge has been constrained to humans, but as digital technology develops and artificial intelligence and machine learning techniques become more mature and integrated with work practices, we are entering the stage where the possibility for more intelligent automation of decision making in asset management increases.

This will be a journey, and while the stage of a fully autonomous critical infrastructure system may never be realised, we are already experiencing the need for more and more hybrid, AI-augmented engineering decision making.

Later in the report an overview of the current state-of-the-art in the application of AI for asset management will be described, but prior to this an exploration of the types of data and information that is available to support asset management, along with the different decisions required are presented.

Sources of data, information and knowledge

There is a wealth of different sources of data which can be used to understand the current health and predict the future health of assets. The main sources of data are as follows:

1) Manufacturing data: Much like any industry, the manufacturing industry is becoming more and more digital. In the context of asset management, it is very rare for the asset owner or operator and the original equipment manufacturer (OEM) to be the same organisation. Asset owners will specify a set of operational criteria and select the most suitable solution from a range of options. There are often many different manufacturing companies in the supply chain, which can be beneficial to prevent common mode failures, but how they interoperate with the rest of the equipment can vary, even if governed by a set of shared standards. Furthermore, the choice of materials and

the manufacturing process itself can have a bearing on the asset life. This is particularly true where the operating regimes and the environmental conditions vary from the standard, anticipated design lifetimes. While valuable, this data is not often transferred between manufacturer and asset owner or operator, as the intellectual property of the manufacturer is tied up in the production of the equipment and they are reticent to share this widely for fear of losing competitive advantage. A further source is data from commissioning tests. This is useful for benchmarking the start of life condition for the asset, and ensures that an asset is accepted as healthy and compliant at the start of life.

2) Location data: While it sounds straightforward, having an asset identifier and an associated geographical location is essential. It provides a strong tie for the asset in the real world with its digital counterpart. While not commonplace, there are certain assets which may be overhauled, possibly uprated and redeployed to other locations within the infrastructure at various points during their operational lifetimes. Examples include, power transformers and pumps, which are high value assets with long lifetimes and suffer from long lead times for new ones, so if part of a network is being reconfigured it can be effective to deploy the existing assets in a new location. Location information is useful for tying to environmental data, such as weather if above ground, and soil composition and co-location with other infrastructure if subsurface.

3) Asset health data: Broadly speaking, asset health data can be separated into two categories. Firstly, inspection data encompasses a broad range of sensor data from visual inspection, to ultrasonic inspection, to chemical analysis of samples. Inspection is either carried out remotely, or if there is a potential risk of harm during inspections, while the asset has been removed from service. Historically, this has been the primary method by which data about the asset is gathered. The second category of data is online condition monitoring data. This data gathering approach has traditionally been associated with protection and control of the asset, but as the ability to cheaply and efficiently gather data, and the increasing prevalence of automated data interpretation techniques, a move towards installing sensors specifically for asset management is becoming more common.

4) Operational data: A significant volume of data to support routine asset operations is gathered. It is essential to understand whether the asset is operating as expected, and many critical infrastructure sectors have control rooms where data about assets across the infrastructure is pooled and decisions are made about interventions to mitigate and clear faults, or to improve the performance of the infrastructure. This data is primarily gathered to support short-term operational decisions, as opposed to condition monitoring data which is used to inform longer timescale asset management decisions.

5) Maintenance records: The outcomes from many inspection and maintenance activities are reported, often using a structured template format. This is particularly prevalent when these activities are outsourced by the asset owner, and acts as a record of the activities undertaken and observations made about the asset. Often these reports summarise the key outcomes of the maintenance, but the associated sensor data is not necessarily recorded.

6) Environmental data: The local environmental conditions in the vicinity of the asset can have an impact on the health and degradation of the asset. For example, the increased salinity of the atmosphere in coastal regions results in corrosion on metal structures. Furthermore, weather and climate can have a significant impact through, for example, localised flooding affecting foundations, and high wind loading in certain locations.

A large volume of data associated with assets is available, both through direct and indirect measurements. For asset management, the current state of the health of an asset must be inferred from sensor data. Furthermore, many critical infrastructure assets have very long design lifetimes, in the order of decades. Gathering and storing data consistently over these long timescales is challenging and as mentioned previously many of the assets do not have digital records of such data stretching to start of life. The ability to capture and store digital data is becoming cheaper and easier, but this leads to a burden in interpretation of the data, to ensure value is derived from it. Many organisations see the use of artificial intelligence as a key technology to support this data interpretation, and recent advances in the field offer significant potential to introduce a step change to the way industrial data is used and managed.

The digitalisation journey
The historical perspective of asset management is very important. Asset management is not a new concept, and large engineering infrastructure has been safely and effectively managed for decades.

Technology has had a significant impact on asset management, but the pace of adoption has been modest. This is largely due to the overarching safety needs, which have been built into operation and maintenance through the use of standards and regulation to define what is acceptable. Early monitoring systems relied on analogue dials and gauges to convey information about the current state of an asset, and this required manually recording in log books. With the introduction of digital sensing technology, this opened up efficiencies in being able to record and assimilate data, and reduced the reliance on engineers being required to be in the physical vicinity of an asset to understand its health. For example, electrical substations used to be manned, but in the late 1990's there was a gradual move towards remote monitoring and operation.

Similarly in the rail industry there was a shift away from manned signal boxes to Rail Operating Centres in the early 2000s. This 20-30 year timescale is relatively short when compared to the age of the physical assets, some of which may be in excess of 70 years old and still in operational use. The pace of change in the digital sector is extremely rapid, and adapting to and capitalising on the benefits of the latest technology is a challenge, particularly in the slow-moving world of critical infrastructure.

Data integrity

In the context of asset management, there is scope for human-performed tasks to be supported and potentially automated by artificial intelligence. Prior to utilising the data to make any asset management decisions (either human driven or AI-enabled), it is important that the data is curated, cleaned and organised in a manner to facilitate interpretation at scale.

Attempting to approach this by building a single, centralised, "data lake" approach is challenging, due to the diversity of the data sources (as discussed in part B), often spread across multiple organisations and derived from sensor technologies of mixed vintages. Standards and ontologies have an important role to play here.

Ontologies provide a common standard set of reference terms to allow both the assets and their data to be described and reasoned about, both by teams of humans, but also utilised by computer algorithms and some forms of artificial intelligence. Within the AI community there are tensions between purely data-driven approaches and symbolic, or knowledge driven approaches. Data-driven approaches can be remarkably effective for some tasks, such as voice and handwriting recognition, object detection and classification and search and summarisation activities, but often lack the ability to explain how they arrived at an answer, or how the complex algorithms actually work.

While in non-safety related applications this may be acceptable, there is a much higher set of consequences for incorrect outputs in applications to safety critical infrastructure. Furthermore, these data driven approaches require large volumes of balanced data in order to perform effectively. In many critical infrastructure applications, the data sets are heavily imbalanced, with large volumes of healthy data and limited instances of faulty data.

The use of synthetic data to artificially balance the data sets for training data-driven approaches is proposed by some, but others claim that this just shifts the problem of acceptance to the synthetic data. Or in other words, how can one be sure that the synthetically generated data is a true enough representation of the faulty data to be relied upon for decision making with safety critical infrastructure?

Symbolic reasoning approaches provide full transparency of algorithmic decision making processes, but suffer from the challenge of curating the knowledge in a structured and generally reusable form.

Ontologies and knowledge graphs aim to capture and represent knowledge in a form independent of any given implementation, much like delimited file formats (such as comma separated values or tab separated values) coupled with the ASCII standard. These have allowed almost universal interpretability of raw sensor data, with almost every modern data logger offering the ability to export to csv, regardless of whether they have integrated visualisation software interfaces and platforms for interpreting the data as part of associated analytics packages. Another important lever for supporting the consistency of data is through standards.

Standards

The development and revision of traditional engineering standards is a slow process. ISO standards typically require 3 years to be developed [32] as part of a multi-stakeholder process, consulting with recognised experts, and the time between successive revisions of standards often significantly exceeds this [33].

However, recent digital innovation has been shown to move more quickly. New software tools are developed following agile methodologies intended to reach markets as quickly as possible, with release cycles measured in weeks or months, rather than years. Machine learning frameworks evolve continuously [34] [35] [36] [37], with major advances occurring more frequently. The result is a growing mismatch between the pace of technological capability advancement and the speed at which standards can be developed and adopted [38].

In response to this, engineering standards may need to evolve to become more agile. Their guidance is updated with too low a resolution to provide guidance on novel methods of data analysis. Given the perceived risks associated with deviation from standards, this lag could inhibit the adoption of new approaches, with the potential to boost infrastructure performance, cost-effectiveness and sustainability.

Learning from (where appropriate) the success of open-source software development projects, engineering standards could become:
– Community driven
– Compatible with continuous integration and testing
– Open-sourced, with respect to data, and models

Successful future engineering standards will need to allow for existing (demonstrably safe) practices, but also for new digital capabilities, where appropriate. This may require new collaborative research, such as joint industry projects, or partnerships with academia.

To build truly resilient critical infrastructure, data and disciplinary silos may need to be broken. Open-science inspired standards and shared data repositories could foster powerful cross-sector learning – enabling, for instance, the application of process engineering's hazard identification to structural safety. Systems-level data collection and analysis has historically been bottlenecked by computational limitations. As this barrier is reduced, inter-disciplinary benefits may emerge.

Key barriers to modern, open science focussed engineering standards may include:
– Institutional inertia
– Liability and legal implications, and regulatory integration
– Maintaining quality assurance and governance, when requesting feedback from a wider community
– A skills and competence gap

The future of CNI standards lies in developing frameworks that can accommodate both the rigour required for safety-critical systems, and the agility necessary to keep pace with technological innovation. The transformation of infrastructure standards represents a critical enabler for realising the full potential of data analytics in critical infrastructure. Success will require coordinated effort across standards organisations, infrastructure operators, technology providers, and regulatory bodies, to develop new approaches that maintain safety and reliability while enabling rapid innovation. This case is made formally in [19].

AI applications in asset management

The term AI was coined by Prof. John McCarthy in the 1950's as a new field of academic study concerned with making computers think like humans. It is also used to describe the systems which exhibit this intelligence. While there is no single agreed definition of artificial intelligence, for the purposes of this report, the Cambridge dictionary defines artificial intelligence as [39]:

'A particular computer system or machine that has some of the qualities that the human brain has, such as the ability to interpret and produce language in a way that seems human, recognize or create images, solve problems, and learn from data supplied to it.'

The field of AI has made incredible advances over the years since its inception, with a myriad of approaches, techniques and algorithms aimed at tackling the range of complex activities that humans are able to achieve. AI is inextricably linked to the environment it operates in, through related fields of sensor technology which convert measurands from the physical to the digital world, and robotics and actuators, which allow outputs from digital algorithms to affect objects in the physical world. Furthermore, through the encoding of information and knowledge in the form of text, images and videos this allows information to be digitally stored and conveyed between people, and increasingly with computer algorithms and AI systems.

This report does not intend to provide a summary of a large variety of tools and techniques which sit under the broad umbrella definition of AI. Instead, this will be explored and articulated by considering the tasks that the AI can be deployed to support within asset management of critical infrastructure.

Once the data is structured and accessible from a single point of access, this opens up the possibility for applying a broad range of AI-based and data analytics techniques for decision support in critical infrastructure asset management.

Common opportunities include:

1) Data visualisation: While not strictly AI, data visualisation can be a powerful way of building confidence with the data user, particularly if the inputs to black box, data-driven techniques can be visualised. The common perspectives of the data should be considered:

a) Spatial, where relationships between the physical location of the assets, or relationships between components within a system can be explored

b) Temporal, where relative changes over time can be visualised

c) Statistical, where general spreads and trends of populations of data can be examined, and normal behaviour can be defined and modelled

d) Procedural, or process driven, or physics informed perspectives where cause and effect relationships can be explored

All these lead to a deeper understanding of the data which in turn leads to an understanding of how and why an asset behaves and degrades, and feeds directly into the other applications below.

2) Anomaly detection: Models of normal behaviour can be derived from data and used to interrogate new, previously unseen data to determine whether it fits the normal, or expected model based on prior experience. If it does not conform to the understanding of what has been observed previously, it's flagged as an anomaly. Anomaly does not necessarily mean a fault, but just that it differs from what was expected. In engineering applications anomaly detection is implicit in protection systems. For example, a threshold is hard coded into the settings of an electromechanical relay, to allow a section of circuit to be tripped in the presence of an abnormal signal. Replicating this process in digital data processing is a form of anomaly detection.

3) Classification: As an extension to anomaly detection, which is essentially a binary classifier (normal or abnormal), classification techniques allow unseen data to be categorised as one of many possible classes. An example of this in critical infrastructure might be categories of different types of damage in underground pipes, such as flow assisted corrosion, material fatigue, microbiologically influenced corrosion and defective welds.

4) Prediction: Building or fitting a mathematical model to the data to enable future values of the data, and thus future health or condition of the asset to be estimated. If a projected end of life state is known, then this prediction can be used to estimate remaining useful life of the asset, and schedule appropriate interventions. Quantifying how the degradation of an asset affects its ability to perform its desired function, and articulating the uncertainty involved with these estimations is another important aspect. An example of prediction in critical infrastructure is estimating the leakage rate of Sulphur Hexafluoride gas in circuit breakers.

5) Scheduling: A significant challenge in inspection and maintenance is scheduling resources. Inspection and maintenance activities are still predominantly a resource intensive activity, and though some progress is being made in automating inspection, for example through the use of autonomous drones to gather visual inspection data, there is still a dependence on physical presence by engineers on site.

Furthermore, critical infrastructure covers a wide geographical area, so ensuring that the right resources are at the right location is complex.

6) Work flow automation and documentation: A broad area where existing workflows are based on recording data and information, predominantly in text format, act as a permanent record of decision making and observation. A lot of these practices are rooted in asset management procedures which pre-date digitisation, and in some cases have not been modernised. This can be particularly true of safety related activities which mandate the production of certain types of report to evidence and justify decision making to external stakeholders, such as the regulator. Recent advances in large language models and generative AI has allowed for much smarter search and summarisation capabilities of large corpuses of text. While these systems are good at conveying the information in a manner which is easily digestible by humans, whether there is underlying understanding beyond the structure of how the information is conveyed is open to debate.

The field of AI is evolving rapidly, with significant investment and uptake across many sectors. However, AI is a loosely defined term and can mean different things to different people, so it is important to be able to define and articulate the human-centred activities the AI is intended to support otherwise there is the risk of investing in and deploying the wrong tool for the job. This will necessitate decisions makers to not only be experts in the appropriate technical discipline, but also be well versed in the relative merits of the different AI algorithms and approaches.

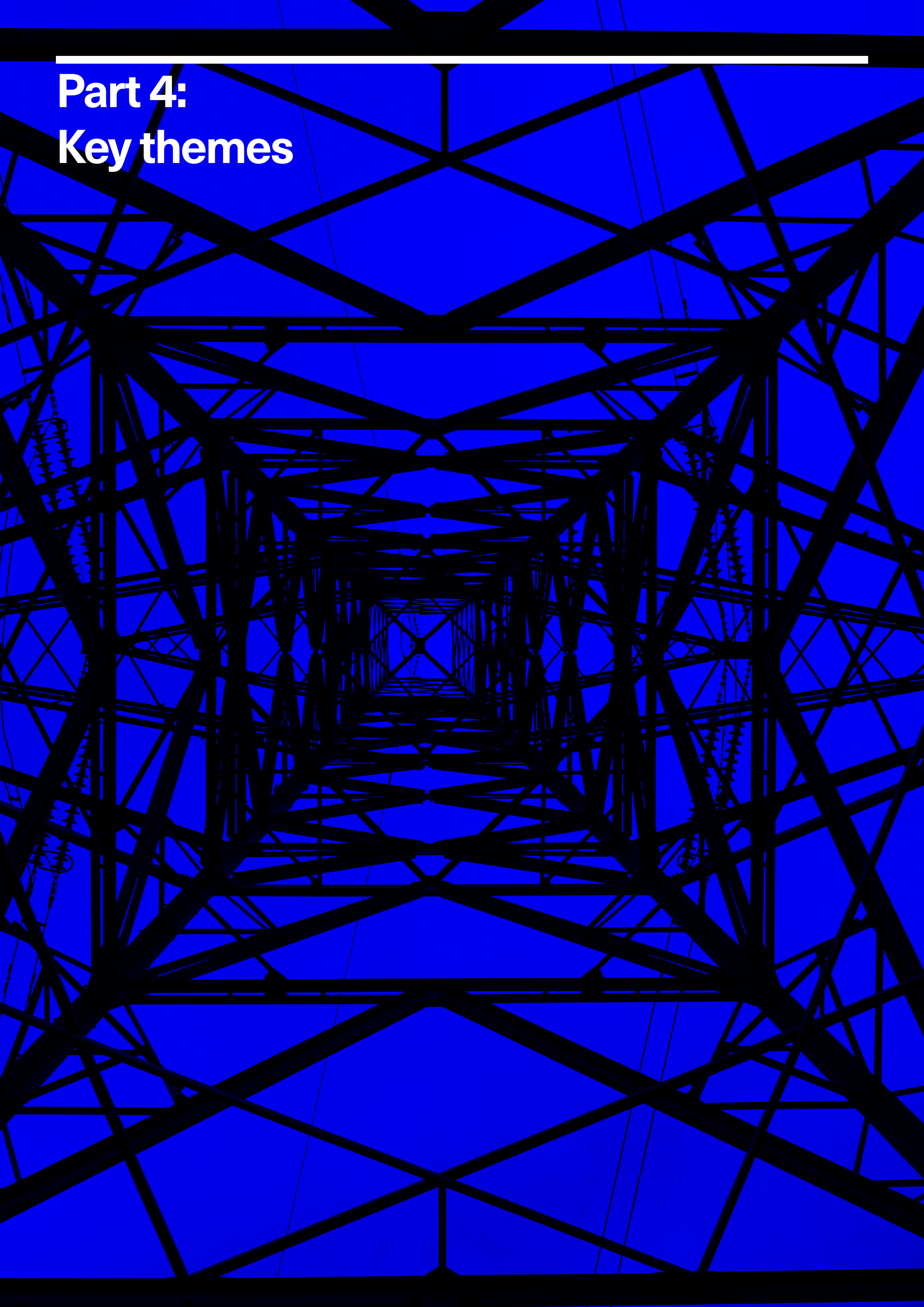
Section summary

Data (raw observations) and the associated information content has long been crucial in supporting asset management decision-making. Various sources, including manufacturing processes, operation and maintenance records, and environmental data, contribute to understanding asset condition and health, and predicting future needs. The increasing availability of digital data, coupled with advancements in analysis capabilities, offers potential for more system-level inferences and optimisations.

There is a fundamental tension between data-driven approaches (which lack explainability but scale well with large datasets) and symbolic reasoning approaches (which provide transparency but require structured knowledge curation). Another significant challenge identified is the mismatch between rapid technological innovation and slow-moving engineering standards, with engineering standards taking years to develop while complex modelling frameworks evolve continuously. More agile and open standards, if developed, could support the safe and appropriate implementations in CNI. This will require careful attention to data integrity, and risk evaluation for safety-critical applications, which fundamentally differ from other sectors.

This transformation is crucial for realising the full potential of data analytics in critical infrastructure. Success will require coordinated effort across standards organisations, infrastructure operators, technology providers, and regulatory bodies to develop new approaches that maintain safety and reliability while enabling rapid innovation.

Part 4:
Key themes



Three common, cross cutting themes emerged from the workshops and associated studies. These themes of: stimulating innovation, the tension behind collaboration and independence, and ensuring the persistence of data, information, and knowledge throughout the asset lifecycle, are discussed herein.

Stimulating innovation

In asset management of critical infrastructure, particularly with the long lifetimes of assets involved, realising the return on investment of introducing new approaches can be challenging. In addition, maintaining and evidencing high levels of reliability over sustained periods where the probability of failure is low, is difficult. By preventing the asset from ever entering a failed state, this also removes the evidence for the upper threshold of acceptable operation. With measures of economic success biased towards short term performance this introduces a barrier to innovation. In regulated environments this has been recognised, and there are both regulator-backed and government-funded initiatives in place to drive innovation into these sectors. Two such regulatory examples are Ofgem's Strategic Innovation Fund (SIF) [40], and Ofwat's Breakthough challenges [41]. An example of a government-driven stimulus was the UKRI's "Innovation in Railway Construction" competition [42].

These are large, national initiatives which generate significant volumes of projects, and increasingly a number of these are focused on the use of data analytics and AI to improve management of assets in these sectors. Some examples which innovate in the digital space are provided in table 4.

This illustrates a level of maturity which is moving the use of AI-based approaches from academia into industry, and business as usual.

There still remains challenges for the adoption of AI approaches in safety critical infrastructure, but even one of the most conservative sectors, nuclear, is seeing a willingness to consider the deployment of AI-based tools [43] [44].

Tension between collaboration and independence

Participants broadly agreed that enhanced collaboration both within and between critical infrastructure sectors could deliver considerable benefits. For example, there was broad agreement that standardisation of data could help break down existing silos, foster interoperability, allow the exploitation of richer datasets, and enable assets to "talk" to each other. A recurring theme was the need for greater third-party asset visibility: one of the asset operators described how, as a 'third party' to many projects, it must expend substantial effort to ascertain whether proposed works will impact its infrastructure and would welcome integrated decision-support tools.

Scheme	Title	Industry	Description
Ofgem SIF	Digital Platform for Leakage Analytics (DPLA)	Gas	Data-driven leakage modelling for proactive leak detection. This includes the rollout of new sensing technologies alongside the implementation of new analytics. Reducing reliance on in-field solutions and creating a more centralised solution.
Ofgem SIF	Intelligent Gas Grid	Gas	Development of an AI tool to use data collected from the network alongside weather data to detect anomalies in gas transportation such as water ingress, gas escapes and malfunctioning governors. Moving away from manual diagnostics to faster remote or automated diagnosis and resolution.
Ofgem SIF	SF6 Whole Life Strategy	Electricity	Project includes developing asset leakage prediction tools using data from existing and new gas density sensors. Also extends to investigating retrofilling and disposal of SF6 switchgear and the in-service behaviour of alternative gas blends looking at the whole lifecycle of switchgear operation and maintenance.
Ofgem SIF	CReDo+ Climate Resilience Demonstrator	Electricity	Development of a digital twin and data sharing platform to enhance resilience investing, planning and reporting. Plan to scale across electricity, gas, water and telecoms sectors to understand interdependencies and build whole system climate resilience.
OFWAT	Safe Smart Systems	Water	The project will use artificial intelligence and mathematical optimisation to improve long-term operational resilience in the face of climate change and rapid population growth. It will identify, predict, and manage vulnerabilities to reduce leakage, interruptions, and pressure issues across the whole water cycle. Safe Smart Systems focuses on the first steps to achieve autonomous control in water systems across the UK.

Table 4: Examples of innovation projects in critical infrastructure

Scheme	Title	Industry	Description
OFWAT	Unlocking Digital Twins	Water	Water companies use these digital twins to unlock new data-driven innovations, which can improve their services and how they manage their networks for the benefit of customers. More companies are using digital twins but there is no agreed standard model and process for creating them, which has the potential to create inefficiencies, reduce the value delivered by these systems and increase costs.
OFWAT	Pipebots for rising mains	Water	Feasibility of developing an 'in-pipe' live rising main condition inspection tool by harnessing the expertise of academia and industry beyond the water sector. The scope of the project was to design, build and test a novel sensor inspection system, mounted on a robotic platform.
GCRE	NAUTILUS Novel Asset Management Twin and Instrumentation System - Feasibility Study	Rail	The NAUTILUS project focuses on developing a novel asset management twin and instrumentation system for the Global Centre of Rail Excellence in South Wales. Nautilus will allow new concept products to be tested virtually to rectify any obvious design problems before they are tested physically on the GCRE equipment, model potential future scenarios, and provide digital feedback on any 'real-world' design changes.
GCRE	Attaining Ubiquitous Railway Analysis (AURA)	Rail	Using optical fibres already installed in the railway aims to rollout three uses: (1) Intruder detection (2) Rolling stock arrival time estimation (3) Detecting and tracking maintenance equipment. Aims to enhance safety whilst improving operational and maintenance efficiency.

Conversely, a rail operator noted that local planning authorities are under no obligation to consult it when approving developments, heightening the risk of inadvertent encroachment on rail assets. Moreover, companies currently employ divergent terminologies for internally identified failures, such as their use of the Apparatus Operational Restriction (AOR) label. Greater collaboration would help to harmonise these standards, avoid interface issues and streamline maintenance. Furthermore, greater collaboration was also perceived as essential to ensuring a System of Systems mentality. In practice, this means sharing information across the CNI landscape in a form that is truly actionable. As one participant explained, “we currently use the data that we have, rather than the data that we need to make an optimal decision.”

Greater collaboration was also seen as essential for establishing trust in advanced analytics and AI adds another layer of complexity: if a model has been developed by an external contractor, how can its outputs be independently assured when the supplier retains ownership of the underlying data? Participants argued for clearly specified ontologies, rules and standards, to underpin AI deployment, and for data-capture protocols that go beyond recording asset state to include the parameters that caused a failure. Improved data collection methods in the wake of incidents, together with mechanisms to scale insights from smaller operators to the largest network owners, would help to diffuse pockets of good practice and foster collective learning.

Despite the numerous opportunities identified, participants frequently alluded to tensions that hinder the adoption of collaborative practices:

1) Standards: Under the Electronic Communications Code (Conditions and Restrictions) Regulations 2003, ‘Code Operators’ licensed by Ofcom are expected to share infrastructure “where practicable”. In reality, the decision rests with the infrastructure owner and there is no requirement to justify a refusal to Ofcom or planning authorities, rendering the obligation more encouragement than compulsion. This mirrors broader gaps in understanding interdependencies between sectors. The Institution for Engineering and Technology (IET) has observed that BT Openreach’s phase-out of PSTN services, critical for gas and energy network telemetry, proceeded with little appreciation of downstream applications, since Openreach has minimal direct contact with end users. The arm’s-length market structure meant that interdependencies and risk scenarios were not captured during planning or risk-assessment stages. Legal constraints on data sharing, including ambiguities over retention periods and release timetables, hinder transparency.

2) The separation of design and build contracts, often leads to cost overruns and blurred accountability. High Speed 2 (HS2) is currently trialling an integrated approach to design and build contracts, showing promise of this type of approach.
3) Failure data is treated as commercially sensitive, shared only when mandated by regulators or when safety imperatives override reticence, perpetuating a

blame culture unfamiliar in sectors such as aerospace.

4) In regulated industries there is a prevailing “if it’s not broken, don’t fix it” ethos, and the perception that data equates to power fuels a reluctance to share. Questions over liability were also raised: for instance, if an AI system recommends that a train proceed when the track is obstructed.

Telecommunications firms illustrate both the promise and pitfalls of cross-sector sharing. It is commonplace to deploy fibre alongside earthing wires on steel pylons, yet health and safety concerns around high-voltage equipment can inhibit access. The unbundling of the electricity system has led to fibre spin-outs such as Energis (formerly part of National Grid) and Neos Networks (from SSE), demonstrating the inherent links between CNI sectors. Water companies have also partnered with fibre providers to share excavation costs and to harness fibre for leak detection, the Department for Science, Innovation and Technology’s (DSIT) Fibre in Sewers (FiS) and the later Fibre in Water (FiW), pilot schemes further exemplify regulatory innovation. Likewise, electricity and rail operators are collaborating on flexible multi-energy hubs, with plans for up to 2,500 sites across the UK rail network.

Despite these challenges, some initiatives demonstrate the potential of collective learning. The National Equipment Defect Reporting Scheme (NEDeRS) obliges operators to share Suspension of Operational Practice (SOP) notices and Dangerous Incident Notifications (DINs) among registered asset owners, albeit within a closed community. Debates continue over what constitutes an ‘innocuous’ failure versus one deserving NEDeRS notification, yet the scheme, which is maintained by the Energy Networks Association, provides a valuable forum for knowledge exchange.

Ensuring the persistence of data, information, and knowledge throughout the asset lifecycle

One of the defining characteristics of critical infrastructure is its longevity. Assets are designed to endure for many decades, often outlasting the people who conceived, commissioned and maintain them. Over that operational lifetime, ownership frequently changes hands, regulations evolve, personnel come and go, and technologies advance. At each juncture there is a real risk that vital Data, Information and Knowledge (DI&K), whether explicitly recorded or held tacitly in people’s heads, may be lost, degraded or rendered unusable by future generations.

With regard to explicit DI&K, participants highlighted several familiar challenges. Data gathered at the design or construction stage is sometimes difficult to locate or interpret years later: paper-based records, bespoke legacy formats and ad-hoc storage practices can frustrate efforts to extract value from historical files. As one contributor observed, when an asset approaches end of life it is essential to ask whether the “beginning-of-life” information is complete, and whether crucial contextual details of its original usage survive. Without certainty about initial conditions, decisions on replacement or refurbishment become guesswork, as illustrated by debates over the fate of ageing sea groynes. The group also noted a paradox of simultaneous data scarcity and excess. While vast volumes of monitoring data may be collected, often simply “because we feel we should”, relatively little is subjected to asset-management-relevant analysis or live diagnosis. High-consequence, low-frequency events complicate statistical training unless synthetic data are introduced, raising questions of validity. Moreover, boundary issues persist around ownership of diagnostic learning: does the asset owner, the monitoring vendor or a contractor hold the intellectual property on closed-loop insights? Even where regulations mandate digital-format submissions (HS2’s data legacy hub being a case in point) owners sometimes discover too late that their records arrive in the wrong schema, forcing costly re-instrumentation or data-repurposing to support future asset management workflows.

Implicit DI&K (expert intuition, hands-on know-how and unspoken judgement) proved no less challenging. For example, a transformer that cracks and pops differently will capture a technician’s attention, but they will know the transformer is not necessarily faulty. Similarly, an experienced technician will be able to “intuit” the subtle signs of a failing water pump. Such skills develop over many years, yet are rarely codified; when experienced staff retire, their accumulated insights risk vanishing.

The group stressed the importance of embedding expert knowledge through structured training programmes, mentoring schemes and, where feasible, codification in decision support tools. Only by clarifying who “owns” that learning (e.g. defining IP boundaries and feedback loops) can the sector hope to sustain the human judgement essential to safe, resilient operation of long-lived infrastructure.

The advent of increasingly AI-enhanced decision-making processes will require the robust codification of explicit and implicit DI&K such that these can be effectively implemented in data-driven and symbolic AI decision-making workflows. Recent advances in multimodal AI could enhance the capture and codification of implicit expert knowledge, as they more closely resemble human forms of knowledge representation. On the other hand, neuro-symbolic approaches could play a crucial role in converting abstract knowledge representations into actionable and explainable rules and routines to be used in the context of AI-enhanced CNI decision-making processes.

Part 5:
Conclusions and recommendations

The aim of this report was to address four questions related to data and information raised by the original study into approaches for ensuring safety of critical infrastructure. This report has explored these issues through the workshops with gathered expertise across the critical infrastructure sectors, combined with a desktop study to further refine the responses. A response to each of the four questions is provided, followed by a roadmap of recommendations for future activities.

What level of predictability (for different asset types and systems) do we need in our infrastructure?

Unfortunately, there isn't a straightforward answer to this. The research conducted in this study shows it largely depends on the criticality of the asset, the appetite for risk, the consequences of failure balanced against the economic argument. Adopting an ALARP (As Low As Reasonably Practicable) approach to mitigate uncertainty in critical infrastructure is a sensible approach, but a balance must be struck between affordability and risk. More recently, with the drive towards net-zero, sustainability has become a third, competing consideration.

Though the question is posed to consider different asset types, the crucial element is to consider the critical infrastructure as a whole, and consequently consider adopting a systems, or systems of systems approach.

What does codification of predictability mean for sourcing and prioritising data needed for decision making (rather than the data which is nice to have)?

Somewhat related to the previous question, this question seeks to prioritise and rank different data sources. As explored through this report, there are a myriad of different data sources and types, coupled with a broad range of different types of decisions to be made at various stages of asset management. Prioritising will come down to the criticality of the asset and the cost associated with capturing and storing the data.

The key takeaway is that the perspective should be starting with the decision that needs to be made, and then prescribing what data is required to make the decision, rather than taking a data-first approach. This latter perspective is arguably more commonplace in "digital first" industries, which is where significant advances in data science and AI has been made as opposed to critical infrastructure which pre-dates digitalisation, and has a long pedigree of safe and effective operation. Significant precaution should be taken if technologies are being forced onto operations where there is not the evidenced and justified reason for doing so. This is recognised within the sectors, and schemes for introducing innovation in a controlled manner are available and proving effective. There is argument that this could be accelerated, but the pace of change in digital technology vastly exceeds the relatively slow-paced evolution of critical infrastructure.

Which data sets, when aggregated, could provide game-changing insights for the design and maintenance of critical infrastructure?

Data aggregation by itself is meaningless, without the associated information about what the data represents and what decisions are being supported in design and maintenance. What is crucial is the provenance and integrity of the data, and that the data is representative of all expected states of the asset. Much like other domains, where non-representative and imbalanced data are fed into the training of machine learning algorithms, the generated output can suffer from bias. Therefore, traceability and provenance of the data is a relatively mundane but crucial aspect of the use of data. With the increasing ease that data can be captured and stored, it is important to ensure that only the relevant data, and its associated context, is provided to the decision maker to prevent data overload.

What type of safe places and safe mechanisms do we need to liberalise data sharing and minimise data protectionism?

There is a tension between the desire to make data open and available to all to support innovation and translation of data-driven analytics techniques in the sector, and the potential for misuse of this data to cause harm and disruption to critical services and wider society. In other sectors a couple of solutions exist to support innovation while protecting data. Firstly, the anonymisation of data is one approach widely adopted. This works well with focused, relatively simple data sets with a single purpose, but difficulties arise when dealing with large, interconnected data sets, where aggregating the data allows relationships and underlying meaning of anonymised data to be inferred from the inherent relationships in the dataset as a whole. A second possible solution which is gaining traction is the use of synthetic data as a surrogate for real data, and to improve balancing of data sets where bias exists.

However, adoption of this approach is still at an early stage, and with safety related applications there remains the question of whether the synthetic data is suitably representative. In terms of collating and providing access to data, the National Underground Asset Register (NUAR) project demonstrates how a phased approach to building a shared data repository, initially with a small subset of users with a specific objective, can be expanded and widened to scale to a broader range of users [18]. This approach has been demonstrated on data which does not require a rigorous security model in place, but could act as a suitable test case should enhanced security for certain types of data be required.

Roadmap towards data-augmented management of critical national infrastructure

To consolidate the outcome of this research, a roadmap has been developed which articulates a series of recommendations, providing specific action points which should be undertaken to facilitate the move towards a data-centric approach to asset management of critical national infrastructure.

Four recommendation areas are identified, and a combination of short, medium and long-term actions is presented.

The first area is focussed on improving the collection and management of digital data, information and knowledge about an asset during inspections.

The second area considers the retention and management of the data in a digital form.

The third area is focused on understanding the degradation of assets and making better predictions about future condition.

The fourth area is a systems-level perspective where asset management decisions and made in the context of operational decision making and optimisation. It is recognised that some organisations and some sectors are already someway down the path.

Recommendation 1: Improved capture of data and information at point of inspection

Action: Short term (1-2 years):

Recommendation 1:
Improved capture of data and information at point of inspection

→ Asset owners should place increased value on the through-life value of the data associated with their assets. The point of capture offers the greatest opportunity to maximise the translation of the physical observations into digital form.

Critical infrastructure exists in the physical world, and inspection of assets represents a key boundary where information about the physical asset health is captured and enters the digital world. Closing the gap between the physical and digital worlds is where immediate gains can be made.

Short term (1-2 years): Provide a suite of tools to facilitate the capture of raw data about the condition of the asset, including condition assessment and mitigating actions taken during periodic inspections. This should be done with cognisance of potential future use of this information for training and supporting AI-based decision support systems. Technologies such as ontologies and knowledge graphs are mature, effective ways of explicitly adding structure to the captured data and information, but are often not yet explicitly embedded within inspection procedures. In addition, tools should be developed to ease this capture in the field in real time, not retrospectively through paperwork and documentation after the inspection has been completed. Voice recognition could be used to capture observations and labels, particularly when combined with image and video data.

A richer set of data could be subsequently used to build up a more granular and representative set of condition ratings, which are typically on a very coarse 5-point scale (from no deterioration to urgent action required) at best. Furthermore, these scales can be subjective and based on the individual inspector's experience, but by gathering and collating the raw data, this can be

Action: Medium term (3-5 years):

As a sector, more work is required to ensure that explicability of outputs from any data analytic support tool is mandated. Justification of decisions is key, particularly where safety is paramount such as in CNI, and this should be a requirement for any AI-based solution.

used to ensure consistency across a body of inspectors. The key for successful adoption of this approach is to ensure that the inspector is not required to do anything additional to the manual reporting process currently undertaken, with little to no additional training required. If reducing the reporting burden through automation can be achieved, then this can ease the adoption process.

Furthermore, emphasising the importance of gathering good data from the outset - by demonstrating how good data capture and labelling can lead to better trained predictive models - can also support the adoption of new processes at the data gathering stage.

Medium term (3-5 years): Where an estimate of the condition of an asset exists prior to inspection, this can be used to provide the inspector with the correct tools to undertake some maintenance and repair activities while the asset is being inspected. There is also the opportunity for a direct comparison to be made between the predicted and as found condition of the asset, though this is rarely done. Closing this feedback loop would allow the opportunity for both a) better understanding of why the prediction may have been wrong and offer an opportunity for making improvements to the predictive model and b) understanding where else this sub-optimal predictive model had been applied, and therefore which predictions may be less reliable than first anticipated. For part a) to be achievable, there needs to be a degree of explicability built into the predictive models, which can be challenging for commonplace black box approaches which have found success in other, less safety critical applications such as voice and handwriting recognition. Physics-informed machine learning and neuro-symbolic approaches are showing promise in this respect, but are not yet as widely adopted and still require further research and development.

Recommendation 2: Improve management and retention of digital data, information and knowledge

Action: Short term (1-2 years):

Recommendation 2:
Improve management and retention of digital data, information and knowledge

Regulation and standards have a strong role to play here to encourage the adoption and interoperability of data and analytics.

Create a hub to promote cross-sharing across national initiatives of best practices for ensuring the persistence of data, information, and knowledge.

Capturing and storing asset health information digitally is now commonplace and is routinely associated with new assets. However, much of the CNI is legacy, pre-dating modern data capture and storage techniques. Providing a digital means to assemble and track the condition of assets is essential to underpin the application of any modern data analytics.

Short term (1-2 years): Create an asset register structure where each asset has a unique digital identifier and corresponds to the asset in the physical world. Associated with this, implement a common language, or ontology to describe the assets digitally. This ontology should allow the data to be described digitally, irrespective of the storage mechanism or analytics tool used to assess the data. It should also be able to store the results of analysis, along with details of the tools, techniques and individuals who produced the results.

Short term (1-2 years): Creation of a hub to promote cross-sharing across national initiatives of best practices for ensuring the persistence of data, information, and knowledge. The recently established Regulatory Innovation Office within DSIT could act as a catalyst to bring together established initiatives to modernise data management of CNI across organisations such as OFGEM, OFWAT, and GCRE. The model followed by such a hub should build upon prior similar knowledge sharing and upskilling initiatives, such as BridgeAI, ADViCE, and the Smart Design Innovation Network.

Action: Medium term (3-5 years):

Modularise and combine the collection and management of data with a set of useful analytics tools. Use flexible architectures to allow you to start small and scale.

There will be a lag as deployed analytics take time to be used and evaluated, particularly given the long lifetimes of some assets. Building in good asset management practice to the models themselves will provide long term benefits.

Action: Long term (5+ years):

Scale and maintain data management architectures. This will include introducing relevant training as new systems and approaches are implemented, but also ensuring that there are mechanisms for future proofing and transferring digital knowledge to new implementations of AI systems.

Medium term (3-5 years): Populate data on an "as needed" basis and develop a suite of simple visualisation tools to support the verification and validation of ingested data, and to identify areas where information is sparse or missing. There is the temptation to start with trying to gather and assimilate all the available asset data into a single monolithic system, before then deciding on which analytics to develop. While this approach can be useful with brand new systems, CNI often contains legacy assets of mixed vintages, which can make ingestion of legacy data challenging. A modular approach could be more suitable, allowing data, and data structures to be added and modified over time, but as mentioned previously, ensuring data quality and integrity should be prioritised over volume.

Medium term (3-5 years): Treat degradation models and analytics as assets themselves. As new observations are made about the health of an asset, evaluate the performance of the modelled predictions. Adopt a software engineering approach to track changes to the model over time, and record which model version was used to support asset management decisions.

Long term (5+ years): Scale and maintain data management architectures. This will include introducing relevant training as new systems and approaches are implemented, but also ensuring that there are mechanisms for future proofing and transferring digital knowledge to new implementations of AI systems.

Recommendation 3: Improve understanding of asset degradation

Action: Short term (1-2 years):

Recommendation 3:
Improve understanding of asset degradation

Predictive models are mature, and should be developed and deployed where possible to evidence their suitability and build confidence in their predictions.

CNI assets typically have long lifetimes with very slow degradation processes. In-service data and post-fault analysis can enhance understanding of asset degradation in-situ, beyond the original design lifetimes of the assets provided by the original equipment manufacturers.

Short term (1-2 years): Data-driven predictive models can be built for single assets where there is a sufficiency of historical data and the degradation mechanisms are well understood. Where appropriate, models should be built and utilised as part of the asset management toolset with appropriate input from both the domain expert and a data analyst (and in some cases, this may be the same individual). This is effective for small, well monitored assets, but can be challenging to scale.

Medium term (3-5 years): Some analytic models require large volumes of data to train, but in asset management applications data tends to be imbalanced, with large volumes of normal behaviour, but relatively little data covering degraded and faulty conditions. The use of synthetic data (where a simulator is used to generate representative data) is gaining traction as a method to redress this imbalance. However, within safety related environments such as CNI where a high degree of trust is required in the outputs, there is concern that the use of synthetic data shifts the problem from that of gathering enough faulty data, to that of validating a model which generates faulty data, without sufficiency of faulty data to validate the model.

Action: Medium term (3-5 years):

Further research is required to build confidence in the use of synthetic data. Demonstration in one safety-critical domain will help build confidence in the application to other related safety-critical domains.

Further research with practical, real world case studies is needed to realise the potential from transfer learning approaches.

The establishment of a sandpit environment to explore and de-risk the implementation of new technologies and approaches in CNI asset management.

Action: Long term (5+ years):

A programme of fundamental research to foresight the next generation of AI-systems and how the lifecycle of the embedded knowledge which supports (and potentially makes) decisions will be managed.

Medium term (3-5 years): An alternative to generating new training data is to further accelerate transfer learning approaches, where a model is developed on one set of data and then applied to a different, but related application by refining, or fine tuning the model at the final stages with domain specific examples.

Medium term (3-5 years): The establishment of a sandpit environment to explore and de-risk the implementation of new technologies and approaches in CNI asset management. The need for safe, trustworthy, and proven methodologies is imperative, and the establishment of a sandpit environment to explore new asset management technology would allow regulators, operators, and owners to come together, trial, and perfect novel approaches without unnecessarily compromising existing infrastructure. Other industries, such as manufacturing, already have successfully leveraged sandpits and demo spaces to trial new technology and approaches (e.g. Advanced Manufacturing Research Centre (AMRC), National Composites Centre (NCC), Manufacturing Technology Centre (MTC)).

Long term (5+ years): The underlying computational technology which supports modern analytics is rapidly changing, as are the reasoning architectures used in AI. Obsolescence of software is an existing challenge facing many industries, and CNI asset management is particularly prone to this due to the long lifespans and slow degradation timescales of the assets.

While data storage is cheap, the sheer volume of data which can be generated over an asset's lifetime will make it impractical to store. Much like humans use models as a surrogate for the raw data, AI approaches are likely to pivot back to incorporating more symbolic-based forms of knowledge representation. There has been little research undertaken on designing AI systems that will last for 40+ years and almost certainly methods will be required where one AI system can pass its accumulated knowledge onto the next generation of system. In engineering terms, accelerated materials degradation is undertaken to understand how structures and systems will behave after many years use, so what would be the equivalent in computing terms? Can experiments be developed to emulate the accelerated degradation of computing technology?

Recommendation 4: Maximise utilisation of non-asset management data

Recommendation 4:
Maximise utilisation of non-asset management data

Digital data is supporting many aspects of critical national infrastructure beyond just the asset management perspective. Operational data, which describes how an asset is utilised, and environmental data which describes the environment in which the asset operates, can both be rich sources of additional data which can be used to augment asset health understanding and management.

Short term (1-2 years): Explore opportunities where operational data can be used to infer asset condition. This data is likely to already be available, but access to the data and an understanding of the relationship between performance and health could provide a coarse measure of overall asset health.

Action: Short term (1-2 years):

Asset owners should undertake a review of data gathered for operational purposes for opportunities to leverage this to support detailed, but infrequent inspections

Exploration of opportunities for automation of multimodal data insight extraction using LLMs

Action: Medium term (3-5 years):

Further research is required to better understand the environmental conditions in which CNI will operate in the mid- to long-term future, and what impact this will have on standard asset management practices.

Short term (1-2 years): LLMs have proven highly capable at shaping, summarising and extracting insights from multimodal data. Extracting insights from thousands of CNI assets manually will be economically and cognitively infeasible, so automation will be required. LLMs, and multiagent LLM systems have already proven useful in automating data analysis, but there are specific safety and security challenges in CNI asset management that must be addressed for their successful implementation.

Medium term (3-5 years): Increase understanding of how environmental conditions and operating duty cycles can impact degradation, which in turn will feed into more complex models of degradation in recommendation 3. Specifically, as the environment changes due to climate change, the associated degradation stressors such as wind loading, moisture and humidity levels, and ambient temperature will all impact asset degradation.

Conclusions and recommendations

A multi-year roadmap and series of recommendations has been presented, and these are agnostic to any given sector. In terms of urgency, the biggest issue to be addressed is bringing the physical world into a digital representation, and making the digital world the first point of contact for understanding asset health. The sectors which have seen the greatest successes in realising the benefits of AI are those where the user is required to interact in the digital world. Born digital sectors, such as social media and gaming are very successful examples. Sectors which had physical aspects such as retail, have adapted and changed significantly, but the tasks being supported (such as search, transactions and information exchange) tend to be simpler than the more complex tasks of asset management (such as health prediction, fault detection). Furthermore, the resulting outcomes (repair and maintenance) requires often complex actions to be performed in the physical world requiring humans to be instructed to perform the tasks. Workflow management systems exist to support these specific tasks, but they are often not joined up digitally to the original analysis tasks which initiate the raising of a work order.

The second most urgent issue is the development of AI and analytics which can effectively explain and justify the decisions being made. Where there is a safety element, such as with CNI the auditability of decision making is essential. Much of the latest successes in AI have been in applications where there is not the same need for decision making transparency. If a voice-controlled music player misinterprets the voice command, or the automated generation of subtitles gets the occasional word wrong there is no significant consequences. However, if an AI algorithm says a pump on a nuclear power plant or a filter on waste water pipe is operating correctly when it isn't, there could be significant repercussions. So, while AI has shown a lot of promise, work is still required to tailor the algorithms to the specific needs of asset management.

One area which could have significant impact is in documentation and reporting of observations. Generative AI and search tools are already demonstrating that efficiency savings can be made in semi-automated documentation generation, but with the caveat that it is used as a tool, not a replacement for the human. Oversight and ownership of the content contained within the report will still lie with an individual, but the production of the documentation can be accelerated. With the human in the loop, this mitigates some of the risks with introducing AI in safety related fields, as ultimate responsibility for veracity still lies with the signatory.

A less urgent but broader observation from the study, is that novel asset management approaches need to be requirements-driven, rather than data-driven. In practice, it was observed that most asset owners and operators suffer from a mixture of data excess, data scarcity, or data waste, usually caused by a prior decision to capture any available data, following the data lake approach prevalent in the mid-2010s, even if it was not clear what that data may or may not be useful for later in the lifecycle. This has led to large amounts of data that are currently difficult to adapt for the requirements of data-supported decision-making processes, which could have been mitigated if a more proactive, requirements-driven data specification and capture had been implemented from the beginning.

Another key takeaway is the gap that currently exists between data-driven and traditional engineering practice. Many participants alluded to the fact that they lack the in-house expertise to process, analyse, and action the data available to them, and frequently need to subcontract with a third-party organisation to conduct these activities. However, this gap has recently begun to close, particularly at the higher education level, with organisations such as The Alan Turing Institute and the Royal Academy of Engineering driving forward the adoption of data-centric engineering education in UK engineering degree curricula.

While this study was focused on the UK, many of these challenges and potential solutions are global in nature. In the USA for example, a major initiative is the National Cyber-informed Engineering Strategy from the U.S. Department of Energy [45] which focuses on challenges to CNI in energy, but explicitly describes how this can be used as a model for other CNI sectors.

Finally, this research has reinforced the paramount importance of encoding human expertise in decision-making processes. Despite the tremendous pace of advances in anomaly detection, physics-informed machine learning, and multimodal generative AI, the ability of a seasoned expert to "intuit" whether a transformer's rattling noise should or should not be a cause for concern remains a fundamental element of the asset management decision-making processes. A statistically triggered flag is meaningless unless actionable meaning is assigned to it. Therefore, if novel data-driven approaches are to truly enhance the asset management decision-making processes, we must deeply examine how current decision-making processes take place, how human judgement and expert knowledge is currently employed, such that we can begin working towards the development of decision-making processes where the harmonious integration of data-driven approaches and expert judgment mutually enhance each other. Recent advances in neuro-symbolic AI have shown promise in helping address this balance, and potential lessons can be learnt from the implementation of AI in medical decision-making processes. One could argue that there is no more critical asset than the human body, so what lessons could we learn from the medical field?

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Appendices

About the authors

Graeme West

Graeme West is Theme Lead for Critical Infrastructure in Data-Centric Engineering at The Alan Turing Institute and a Professor in Electronic and Electrical Engineering at the University of Strathclyde. He completed his BEng degree in Electrical and Mechanical Engineering in 1998 and his PhD in 2003 in Computational Intelligence Methods for Power System Protection Design and Decision Support both at the University of Strathclyde. He is the industrial informatics lead in the Advanced Nuclear Research Centre at Strathclyde and has over 20 years' experience developing and deploying intelligent decision support systems to the power industry, and in particular to through life asset management of nuclear generation assets. Graeme has particular interest in the fields of knowledge engineering, and the capture codification and utilisation of human expertise coupled with the recent advances in machine learning and deep learning.

Francisco Gómez Medina

Francisco is a Research Application Manager in Data-Centric Engineering at The Alan Turing Institute. Before joining the Turing, Francisco completed his PhD at the Institute for Manufacturing, University of Cambridge. His doctoral research focused on understanding the value of Digital Twins in heavy asset industries, such as aerospace and wind renewables, with a focus on organizational learning. He is also a researcher and advisor at CIEMAT, Spain's Centre for Energy, Environmental and Technological Research, focusing on modelling and diagnostics of renewable energy infrastructure. Prior to his PhD, he spent time working at the Bristol Robotics Laboratory and co-founded two startups in the gamified advertising and education sectors. He also holds a MEng degree in Mechanical Engineering from the University of Bristol.

Domenic Di Francesco
Domenic is a Turing Research Fellow at The Alan Turing Institute, and a Visiting Researcher at the University of Cambridge. Domenic is also a chartered mechanical engineer who has previously worked in the energy industry. His PhD investigated the application of partial pooling models in value of information analysis, with application to structural integrity management. Since joining The Alan Turing Institute, he has continued to work on Bayesian decision analysis, system effects, and model risk. He has also contributed to industrial projects with partners including Lloyds Register, the Health & Safety Executive (HSE) and the Cambridgeshire and Peterborough Integrated Care Board (ICB). These projects have focused on guidance for engineers and clinicians on the use of computational statistics, and AI assurance.

Rossita Maclean
Rossita is currently undertaking a PhD at the Queen's University Belfast in the School of Electronics, Electrical engineering, and Computer Science, and recently completed an Internship at The Alan Turing Institute in the Data-Centric Engineering programme. Rossita's PhD is entitled: "Identification and mitigation of Very Low Frequency Oscillations (VLFOs) on the all-island power system". Her current research is looking at PMU data analysis and the comparison of detection methods for VLFOs, taking algorithms and approaches from different industries and applying them to power systems data. Rossita also holds a MEng from Queen's University Belfast, during which she investigated black start solutions using distributed generation for the Northern Irish grid. This included simulation studies using PowerFactory DIGiSilent and culminated in a report with technical and regulatory recommendations.

Workshop participants

The authors gratefully acknowledge the valuable contributions and insightful discussions provided by all participants during the workshops held throughout this project. Their active engagement and diverse perspectives were instrumental in shaping the recommendations presented in this report. Please see the full list of participants below:

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	Julian Roberts	Head of Strategy	Curious Blue Fish
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